Canadian Residential Energy-Use Intensity Benchmarks by Dwelling Type

Establishing empirically validated energy-use intensity (EUI) benchmarks is a critical prerequisite for calibrating and validating building-energy simulation results. To verify the physical plausibility of computational models, simulated EUIs must be cross-referenced with empirical baseline data from official administrative datasets. Natural Resources Canada (NRCan) administers the primary databases capturing these trends: the Survey of Household Energy Use (SHEU) and the Comprehensive Energy Use Database (CEUD).[1]

The primary metric of interest for building performance engineers is secondary energy-use intensity, which represents the energy delivered to the building boundary. This analysis synthesizes these database records for the 2019 calendar year to construct baseline plausibility bands—expressed in both Gigajoules per square meter ($\mathrm{GJ/m^2}$) and kilowatt-hours per square meter ($\mathrm{kWh/m^2}$)—categorized by residential dwelling type.[1]

Empirical EUI Benchmarking Matrix

The following multi-variable table summarizes the national averages, electricity-only components, derived architectural areas, and regional boundaries established by NRCan.[4] The conversions from Gigajoules to kilowatt-hours utilize the standard factor:

$$1\ \mathrm{GJ} = 277.78\ \mathrm{kWh}$$

Residential Energy-Use Intensity Benchmarks (2019 Database Records)

Dwelling Type	Energy Intensity — TOTAL All-Fuels (GJ/m2 and kWh/m2)	Energy Intensity — ELECTRICITY Only (kWh/m2)	Low–High Range Across Provinces (GJ/m2 and kWh/m2)	Source Table, Year, and Database Archive
Single Detached	$0.56\ \mathrm{GJ/m^2}$ ($155.56\ \mathrm{kWh/}$	$57.89\ \mathrm{kWh/m}$ *(Derived; see derivation)* [1]	0.47 **to** $0.59\ \mathrm{GJ/m^2}$ (130.56 to	SHEU 2019 Table 3.3b, Table 3.5b (https://oee.nrcan.gc.ca/corpo

	) [6]		$163.89\ \mathrm{kWh/m^2}$) *(Partial provincial range)* [6]	[rate/statistics/neud/dpa/menus/sheu/2019/tables.cfm](https://oee.nrcan.gc.ca/corporate/statistics/neud/dpa/menus/sheu/2019/tables.cfm)) [3]
Single Attached *(Semi-detached / Row / Townhouse / Duplex)*	$0.52\ \mathrm{GJ/m^2}$ ($144.45\ \mathrm{kWh/m^2}$) [6]	**NOT FOUND** *(Total energy/hh missing from source)* [5]	0.51 **to** $0.52\ \mathrm{GJ/m^2}$ (141.67 to $144.45\ \mathrm{kWh/m^2}$) *(Partial provincial range)* [6]	SHEU 2019 Table 3.3b, Table 3.5b (https://oee.nrcan.gc.ca/corporate/statistics/neud/dpa/menus/sheu/2019/tables.cfm) [3]
Apartment, Low-Rise *(<5 Storeys)*	$0.52\ \mathrm{GJ/m^2}$ ($144.45\ \mathrm{kWh/m^2}$) [6]	**NOT FOUND** *(Total energy/hh missing from source)* [5]	0.40 **to** $0.78\ \mathrm{GJ/m^2}$ (111.11 to $216.67\ \mathrm{kWh/m^2}$) *(Complete regional range)* [6]	SHEU 2019 Table 3.3b, Table 3.5b (https://oee.nrcan.gc.ca/corporate/statistics/neud/dpa/menus/sheu/2019/tables.cfm) [3]
Apartment, High-Rise *(≥5 Storeys)*	**NOT FOUND** *(Total EUI missing from source)* [1]	**NOT FOUND** *(Total EUI missing from source)* [1]	**NOT FOUND** [3]	SHEU 2019 Table 3.3b, Table 3.5b ([https://oee.nrcan.gc.ca/corporate/statistics/neud/dpa/menu

				s/sheu/2019/tables.cfm) [3]
Apartment, Combined *(CEUD Aggregate Class)*	$0.56\ \mathrm{GJ/m^2}$ ($155.56\ \mathrm{kWh/}$) [4]	$84.64\ \mathrm{kWh/m}$ *(Derived; see derivation)* [4]	**NOT FOUND** *(Not published regionally in CEUD)* [7]	CEUD Table 45 (Canada, 2019) (https://oee.nrcan.gc.ca/corporate/statistics/neud/dpa/showTable.cfm?type=CP&sector=res&juris=ca&year=2023&rn=45&page=0) [4]

Note on Combined Apartment Row: Because high-rise apartment EUI data is partially missing due to source truncation, the aggregate "Apartment" category from the CEUD database has been provided to serve as a consolidated comparison baseline for multi-unit modeling.[4]

Mathematical Formulations and Spatial Normalization

Single Detached Dwelling Derivations

Because SHEU 2019 Table 3.5b publishes electricity intensity on a "per household" basis ($45.7\ \mathrm{GJ/household}$) rather than a "per unit area" basis, an empirical heated floor area must be derived to resolve the electricity EUI ($\mathrm{kWh/m^2}$).[5] This is accomplished by dividing the average household total energy consumption by the total energy intensity per square meter [1]:

$$\text{Average Floor Area } (A_{\text{heated}}) = \frac{\text{Total Household Energy Use}}{\text{Total Energy Intensity per Square Meter}}$$

$$A_{\text{heated}} = \frac{122.8\ \mathrm{GJ/household}}{0.56\ \mathrm{GJ/m^2}} = 219.29\ \mathrm{m^2}$$

This derived architectural heated area of $219.29\ \mathrm{m^2}$ is subsequently utilized to solve for the electricity-specific EUI [5]:

$$\text{Electricity EUI} = \frac{\text{Electricity Intensity per Household}}{A_{\text{heated}}}$$

$$\text{Electricity EUI} = \frac{45.7\ \text{GJ/household}}{219.29\ \text{m}^2} = 0.2084\ \text{GJ/m}^2$$

Converting this value to electrical EUI yields:

$$0.2084\ \text{GJ/m}^2 \times\ 277.78\ \text{kWh/GJ} =\ 57.89\ \text{kWh/m}^2$$

Apartment Combined Dwelling Derivations

Within the CEUD framework, apartments are aggregated into a single housing stock class.[4] In 2019, the total floor space of the apartment sector was $507\ \text{million m}^2$ distributed across $4.671\ \text{million households}$, establishing a derived average floor area of [4]:

$$\text{Average Apartment Floor Area} = \frac{507 \times 10^6\ \text{m}^2}{4.671 \times 10^6\ \text{households}} = 108.54\ \text{m}^2\ \text{per household}$$

CEUD Table 45 lists the total electrical consumption of the apartment sector as $154.5\ \text{PJ}$ ($1.545 \times 10^8\ \text{GJ}$).[4] The electricity EUI is calculated directly from the sector totals:

$$\text{Electricity EUI} = \frac{\text{Total Apartment Electricity Use}}{\text{Total Apartment Floor Space}}$$

$$\text{Electricity EUI} = \frac{154.5 \times 10^{15}\ \text{Joules}}{507 \times 10^6\ \text{m}^2} = 0.3047\ \text{GJ/m}^2$$

$$0.3047\ \text{GJ/m}^2 \times\ 277.78\ \text{kWh/GJ} =\ 84.64\ \text{kWh/m}^2$$

This derived intensity translates to an average household electrical draw of $33.08\ \text{GJ/household}$, aligning with the spatial EUI when divided by the derived floor area of $108.54\ \text{m}^2$.[4]

Rationale for Missing Values

The electricity EUIs for Single Attached and Apartment, Low-Rise are designated as "NOT FOUND" because the specific datasets lack a matching total energy use per household ($E_{\text{total, hh}}$) or an explicit floor area.[5] Without a joint record of total energy per household and total energy intensity per square meter, the floor plate area cannot be mathematically derived. Introducing arbitrary assumptions or borrowing floor areas from other vintages would

compromise the scientific integrity of the calibration bands, making "NOT FOUND" the only rigorous designation.
Similarly, high-rise apartment metrics are marked as "NOT FOUND" because Table 3.3b does not provide the corresponding total energy intensity per heated area for this category in the available records, preventing the geometric decoupling of household energy use into spatial intensity.[5]

Socio-Technical and Climatic Drivers of Residential Energy Use

Volumetric Dilution vs. Density Concentration

A comparative evaluation of these benchmarks reveals a significant architectural and thermodynamic phenomenon. When analyzed on a "per household" basis, multi-unit residential buildings (MURBs) demonstrate superior energy efficiency, with high-rise apartments consuming just $39.3\ \mathrm{GJ/household}$ compared to $122.8\ \mathrm{GJ/household}$ for single detached dwellings.[1] However, when normalized by floor area (EUI), this advantage is heavily compressed. Low-rise apartments demonstrate a total EUI of $0.52\ \mathrm{GJ/m^2}$ ($144.45\ \mathrm{kWh/m^2}$), which is nearly identical to single-attached homes ($0.52\ \mathrm{GJ/m^2}$) and only slightly lower than single-detached dwellings ($0.56\ \mathrm{GJ/m^2}$).[6] When looking at the CEUD aggregate apartment class, the total EUI actually matches the single detached EUI at $0.56\ \mathrm{GJ/m^2}$.[4]
This phenomenon occurs because apartments feature higher occupant densities, more continuous mechanical ventilation requirements, and elevated plug loads per square meter. In contrast, while single detached homes have larger exterior thermal envelopes exposed to winter conditions, their vast floor plates dilute their volumetric energy use when normalized per square meter.

The Electrification Dichotomy

The secondary energy mix varies drastically by building type. In single detached dwellings, electricity represents a minority share of total end-use energy (derived at $57.89\ \mathrm{kWh/m^2}$ or 37.2% of total energy).[1] The remainder is dominated by combustion fuels, primarily natural gas utilized for space and water heating.[1]
Conversely, the CEUD apartment sector exhibits a highly electrified profile, with electricity constituting 54.4% of total secondary energy use ($154.5\ \mathrm{PJ}$ out of $284.2\ \mathrm{PJ}$).[4] This results in an electricity EUI of $84.64\ \mathrm{kWh/m^2}$.[4] This high electrical intensity is driven by the prevalence of electric baseboard heating systems in multi-unit buildings, particularly in

Quebec, which consumes 40% of all residential electricity in Canada.[1]

Regional Climatic Gradients

The low-high range compiled across provinces serves as an excellent boundary for simulation calibration. For low-rise apartments, the regional EUI spans from a low of $0.40\ \mathrm{GJ/m^2}$ ($111.11\ \mathrm{kWh/m^2}$) in the Atlantic provinces up to a high of $0.78\ \mathrm{GJ/m^2}$ ($216.67\ \mathrm{kWh/m^2}$) in Alberta.[6]

This variance is caused by two compounding factors:

1. **Climatic Exposure:** Regions like Manitoba and Saskatchewan experience extreme heating degree days (HDDs), forcing higher space-heating thermal demands per square meter ($0.67\ \mathrm{GJ/m^2}$ average).[1]
2. **Infrastructure and Energy Availability:** The Atlantic provinces rely heavily on wood and oil for supplementary heating, which compresses secondary electricity and natural gas intensities, whereas Alberta's residential sector relies almost exclusively on carbon-intensive fossil fuel combustion and space heating systems.[1]

Database Access Paths and Technical Specifications

To verify the validity of these benchmarks, researchers can cross-reference the data points with the following official database parameters:

- **Survey Title:** Survey of Household Energy Use (SHEU) 2019 [1]
- **Co-Publishers:** Natural Resources Canada (NRCan) & Statistics Canada [1]
- **Database Catalog Number:** SHEU-2019 [1]
- **Target Population:** Private Canadian dwellings, excluding households in the territories, on First Nations reserves, and military bases [2]
- **Reference Year:** 2019 [1]
- **Release Year:** 2021 [1]
- **Primary SHEU Table Access Path:**(https://oee.nrcan.gc.ca/corporate/statistics/neud/dpa/menus/sheu/2019/tables.cfm) [3]
- **Primary CEUD Table Access Path:**(https://oee.nrcan.gc.ca/corporate/statistics/neud/dpa/showTable.cfm?type=CP&sector=res&juris=ca&year=2023&rn=45&page=0) [4]

Works cited

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